Mini Project Report

on

Synsera

Submitted for partial fulfillment of award of

BACHELOR OF TECHNOLOGY

degree

In

Computer Science & Engineering

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Dec 2025

STUDENT'S DECLARATION

We hereby declare that the work being presented in this report entitled "Synsera" is an authentic record of our own work carried out under the supervision of Dr. Shanu Sharma.

The matter embodied in this report has not been submitted by us for the award of any other degree.

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CERTIFICATE

This is to certify that project report entitled “Synsera” which is submitted by Akshansh Shekhar, Ayush Goyal and Ayush Kumar Singh in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science & Engineering of Dr. A.P.J. Abdul Kalam Technical University, formerly Uttar Pradesh Technical University is a record of the candidate own work carried out by him/them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree.

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ABSTRACT

In modern organizations, employee well-being and productivity have become critical factors influencing overall performance and sustainability. With the increasing reliance on digital communication, large volumes of textual employee feedback remain underutilized for understanding emotional and psychological states. This project, titled ***Synsera****, aims to address this gap by developing an AI-based system for analyzing employee sentiment and identifying potential burnout risks through text-based inputs.*

The problem addressed in this project is the lack of continuous, automated, and non-intrusive mechanisms to monitor employee emotions and morale in real time. Traditional feedback systems are periodic and manual, making it difficult to detect early warning signs of stress or disengagement.

To investigate this problem, Synsera employs Natural Language Processing and Machine Learning techniques to analyze employee check-ins. The system preprocesses textual data and classifies sentiment using trained models, followed by trend analysis to assess emotional patterns over time. A web-based platform integrates sentiment analysis with dashboards for employees, managers, and HR personnel.

The results demonstrate accurate sentiment classification, effective trend visualization, and reliable detection of burnout indicators under normal operating conditions. The system successfully provides meaningful insights while maintaining data privacy and usability.

In conclusion, the objectives of the project have been achieved. Synsera demonstrates the feasibility of applying AI-driven sentiment analysis for workplace well-being monitoring and provides a strong foundation for future enhancements such as predictive analytics and voice-based emotion detection.

Keywords: Sentiment Analysis, Natural Language Processing, Machine Learning, Employee Well-being, Burnout Detection

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LIST OF SYMBOLS

Symbol	Description
%	Percentage
$>$	Greater than
$<$	Less than
$=$	Equal to
$\rightarrow$	Data flow direction
Σ	Aggregation / Summation
θ	Model parameter
$f(x)$	Sentiment classification function

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
DFD	Data Flow Diagram
ER	Entity Relationship
UML	Unified Modeling Language
JWT	JSON Web Token
API	Application Programming Interface
HR	Human Resources
UI	User Interface
DB	Database

CHAPTER 1

INTRODUCTION

In today's fast-paced professional environment, employee well-being and productivity have emerged as critical factors influencing organizational success. While technological advancements have significantly improved operational efficiency, understanding the emotional and psychological state of employees remains a challenge. Traditional performance evaluation systems largely focus on quantitative metrics and often fail to capture employees' sentiments, stress levels, and overall morale. As a result, organizations struggle to identify early signs of burnout, disengagement, or declining productivity.

With the rapid growth of Artificial Intelligence (AI) and Natural Language Processing (NLP), it has become possible to analyze human emotions and opinions expressed through textual data. Sentiment analysis enables organizations to gain meaningful insights from employee feedback, daily check-ins, and work-related reflections. Motivated by this need, the project **Synsera** has been designed as an AI-based platform to analyze employee sentiments and provide actionable insights for team leads and management.

This chapter introduces the problem domain, highlights the motivation behind the project, defines its objectives and scope, reviews related work in the field, and outlines the organization of the report.

1.1. Problem Introduction

Organizations today rely heavily on teamwork, collaboration, and employee engagement. However, employees often hesitate to openly express stress, dissatisfaction, or emotional challenges, leading to unnoticed burnout and reduced efficiency. Existing feedback mechanisms such as surveys or periodic reviews are infrequent, manual, and lack real-time sentiment evaluation.

The problem addressed in this project is the **lack of an automated, continuous, and intelligent system** that can analyze employee emotions from textual inputs and provide timely insights regarding morale, productivity trends, and burnout risks. There is a need for a system that not only collects employee feedback but also interprets it meaningfully using AI techniques.

Synsera aims to bridge this gap by enabling employees to submit regular textual check-ins describing their tasks, challenges, and feelings, which are then analyzed using NLP-based sentiment analysis techniques.

1.1.1. Motivation

The primary motivation for developing Synsera arises from the increasing importance of mental well-being in workplaces and academic teams. Studies have shown that emotionally satisfied employees are more productive, innovative, and committed to their organizations. However, managers and HR teams often lack tools to continuously monitor emotional health without being intrusive.

Advancements in AI and NLP provide an opportunity to automatically analyze textual data and detect underlying emotional patterns. This project is motivated by the desire to apply these technologies in a practical, socially impactful manner to improve transparency, communication, and overall team well-being.

1.1.2. Project Objective

The main objectives of the Synsera project are as follows:

- To design and develop an AI-based platform for regular employee check-ins.
- To analyze textual inputs using Natural Language Processing techniques.
- To classify employee sentiments into positive, negative, or neutral categories.
- To identify trends related to productivity and burnout risk.
- To generate meaningful insights and summaries for team leads and HR personnel.
- To promote transparency and open communication within teams.

1.1.3. Scope of the Project

The scope of this project includes the development of a full-stack web application that performs sentiment analysis on employee-submitted text data. The system focuses on text-based emotion detection and does not include biometric or physiological data analysis.

Synsera is applicable to:

- Corporate teams
- Startups
- College clubs and project teams
- HR departments for morale tracking

Future enhancements such as voice-based sentiment detection, third-party tool integration, and advanced predictive analytics are beyond the current scope but have been identified for further development.

1.2. Related Previous Work

Significant research has been carried out in the field of sentiment analysis and emotion detection using Artificial Intelligence and Natural Language Processing. Early approaches relied on rule-based and lexicon-based techniques, while later developments introduced machine learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Logistic Regression for sentiment classification.

Recent advancements include deep learning models like Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, which are capable of capturing contextual relationships within text. Sentiment analysis has been widely applied in domains such as product reviews, social media monitoring, customer feedback analysis, and brand reputation management.

However, limited work has focused on applying sentiment analysis specifically to employee well-being and internal productivity monitoring. Synsera builds upon existing research by adapting these techniques to organizational environments and providing actionable insights tailored for team management.

1.3. Organization of the Report.

The remainder of this report is organized as follows:

- **Chapter 2** presents the literature review and discusses existing techniques and methodologies related to sentiment analysis and employee well-being monitoring.
- **Chapter 3** describes the system architecture, tools, technologies, and overall design of the Synsera platform.
- **Chapter 4** explains the implementation details, including data preprocessing, model training, and system integration.
- **Chapter 5** discusses the results obtained, system performance, and analysis of outcomes.
- **Chapter 6** concludes the report and outlines future scope and possible enhancements to the system.

LITERATURE SURVEY

Sentiment analysis has gained significant importance with the rapid growth of digital communication and data-driven decision-making. Researchers have explored various techniques to analyze emotions, opinions, and attitudes expressed in textual data. In organizational environments, sentiment analysis plays a crucial role in understanding employee satisfaction, stress levels, and overall morale. Several Natural Language Processing (NLP) and Machine Learning (ML) techniques have been proposed to classify sentiments and detect emotional patterns from text. This chapter presents a survey of existing techniques, algorithms, and technologies relevant to employee sentiment analysis and productivity monitoring systems.

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1 Sentiment Analysis Using Natural Language Processing

Sentiment analysis is a subfield of Natural Language Processing that focuses on identifying and classifying emotions expressed in text. It aims to determine whether a given piece of text conveys a positive, negative, or neutral sentiment. NLP techniques such as tokenization, part-of-speech tagging, lemmatization, and syntactic parsing are commonly used to preprocess textual data before sentiment classification.

Early NLP-based sentiment analysis systems relied on rule-based and lexicon-based approaches, where predefined dictionaries of positive and negative words were used to compute sentiment scores. Although these methods are simple and interpretable, they struggle with sarcasm, context, and domain-specific language. Despite their limitations, lexicon-based approaches laid the foundation for modern sentiment analysis systems and are still used as baseline models.

Table 1 Sentiment Categories and Examples

Sentiment Type	Description	Example Sentence
Positive	Indicates satisfaction or happiness	“I feel motivated and productive today.”
Negative	Indicates stress or dissatisfaction	“The workload is overwhelming.”
Neutral	No strong emotional tone	“I completed my assigned task.”

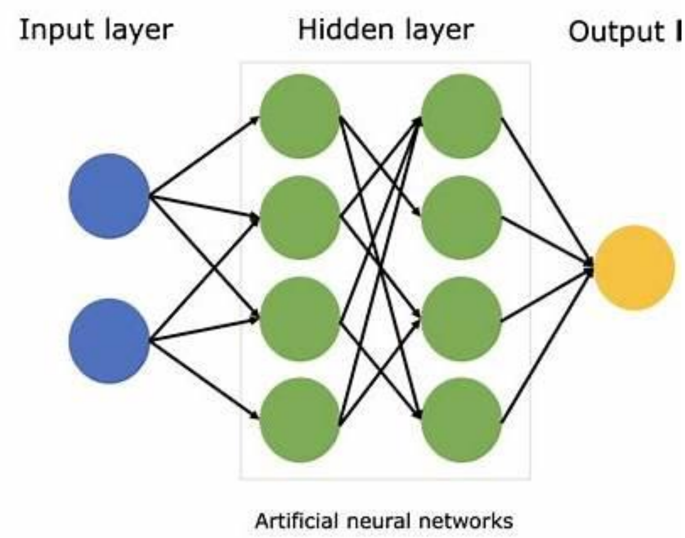

Figure 1: Typical Neural Network for Sentiment Analysis

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2 Machine Learning-Based Sentiment Classification

Machine learning techniques significantly improved the accuracy of sentiment analysis by learning patterns directly from labeled data. In these approaches, text data is converted into numerical features using methods such as Bag of Words (BoW), Term Frequency–Inverse Document Frequency (TF-IDF), or word embeddings. These features are then used to train classification models.

Algorithms such as Naïve Bayes, Logistic Regression, and Support Vector Machines (SVM) are widely used for sentiment classification. Naïve Bayes is computationally efficient and performs well on text classification tasks with large vocabularies. Logistic Regression provides probabilistic outputs and is effective for binary and multi-class sentiment classification. These models are easy to implement and interpret, making them suitable for practical applications such as employee feedback analysis.

Table 2 Comparison of Machine Learning Algorithms

Algorithm	Advantages	Limitations
Naïve Bayes	Fast, simple, works well on text	Assumes feature independence
Logistic Regression	Interpretable, good accuracy	Requires feature engineering
SVM	High accuracy	Computationally expensive

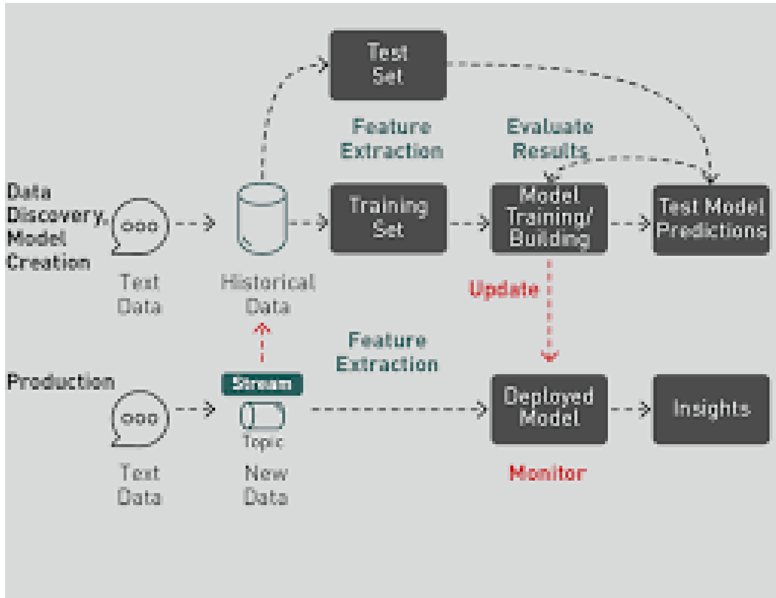

Figure 2: Machine Learning–Based Sentiment Classification Process

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3 Deep Learning Approaches for Contextual Analysis

Deep learning models capture semantic and contextual information from text more effectively than traditional methods. Architectures such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks are commonly used.

A neural network consists of an input layer, one or more hidden layers, and an output layer. These networks learn hierarchical features automatically during training. Deep learning models are effective in handling long sentences, sarcasm, and contextual dependencies.

Table 3 Comparison of CNN and LSTM Models

Model	Description	Strengths	Limitations
CNN	Extracts local features using convolution filters	Fast, good for key phrase detection	Limited long-term context
LSTM	Processes text sequentially using memory cells	Handles long-term dependencies	Computationally expensive

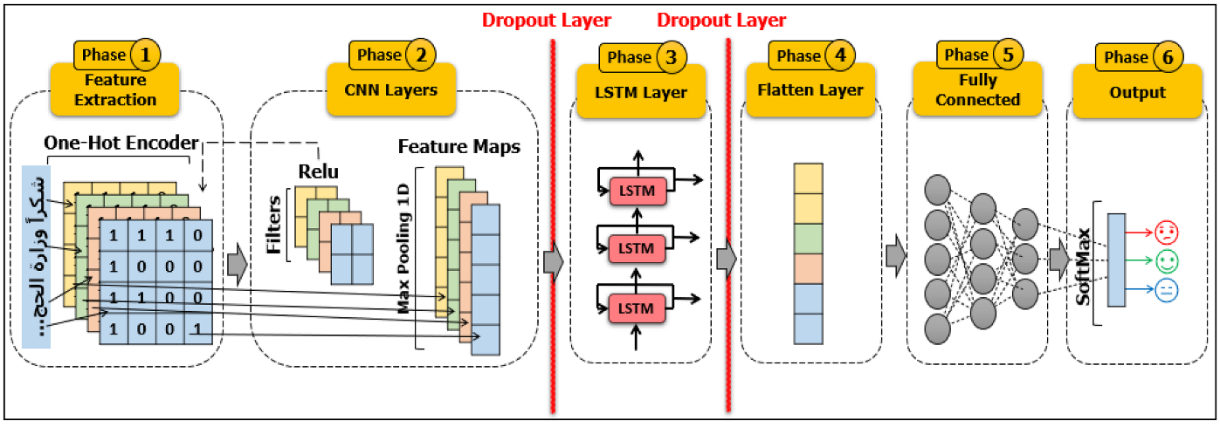

Figure 3 Deep Learning Architecture for Sentiment Analysis

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4 Employee Sentiment and Burnout Analysis

Employee sentiment analysis focuses on evaluating emotional patterns from workplace-related text such as feedback, surveys, and daily check-ins. By tracking sentiment trends over time, organizations can identify early signs of burnout and declining productivity.

Research indicates that negative sentiment trends correlate with increased stress and disengagement. AI-based sentiment monitoring systems help managers take preventive actions such as workload redistribution and team counseling.

Table 4 Indicators of Employee Burnout Based on Sentiment Trends

Indicator	Description	Organizational Impact
Continuous Negative Sentiment	Repeated negative emotional expressions	Increased burnout risk
Declining Sentiment Score	Gradual drop in sentiment over time	Reduced productivity
Emotional Variability	Sudden mood fluctuations	Work instability
Neutral Dominance	Lack of emotional expression	Possible disengagement

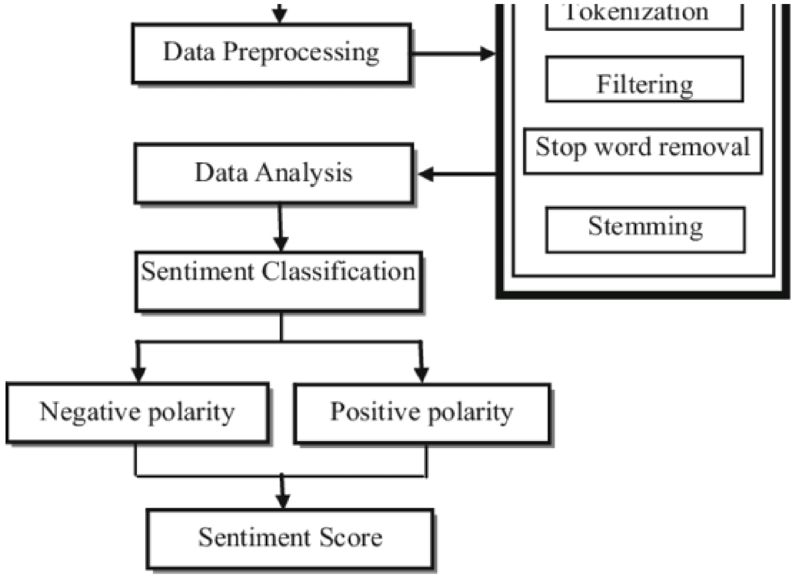

Figure 4 Employee Sentiment and Burnout Detection Workflow

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5 Tools and Technologies for Sentiment Analysis

Several tools support the development of sentiment analysis systems. NLP libraries such as NLTK and spaCy provide preprocessing and linguistic analysis features. Machine learning frameworks like scikit-learn enable model training and evaluation. Web-based platforms integrate AI models with backend services and databases to provide real-time insights.

Machine learning frameworks enable the training and evaluation of sentiment classification models using numerical text representations such as Bag of Words and TF-IDF. For real-time usage, trained models are integrated with backend services through APIs, allowing seamless communication between the user interface and the analysis engine. Databases store employee feedback and sentiment results, while frontend technologies provide dashboards and visualizations to present insights in an understandable manner.

Table 5 Tools Used in Sentiment Analysis Systems

Tool / Technology	Purpose
NLTK	Text preprocessing
spaCy	Advanced NLP processing
scikit-learn	ML model training
Flask / Node.js	Backend APIs
MySQL	Data storage

SUMMARY

This chapter surveyed existing research and technologies related to sentiment analysis and employee well-being monitoring. NLP-based lexicon methods were discussed as foundational approaches, followed by machine learning techniques such as Naïve Bayes and Logistic Regression. Deep learning models were explored for their ability to capture contextual meaning in text. Employee sentiment and burnout analysis techniques were reviewed, emphasizing their importance in modern organizations. Finally, tools and technologies used in implementing sentiment analysis systems were summarized. These studies provide a strong theoretical foundation for the design and implementation of the Synsera platform.

CHAPTER 2

SOFTWARE REQUIREMENT SPECIFICATION

This section describes the general factors that affect the Synsera system and its requirements. It provides a high-level understanding of the product, its operating environment, user characteristics, and constraints. The purpose of this section is to present the system requirements in simple, non-technical language for stakeholders and customers, while preparing the ground for the detailed technical specifications described in later sections.

2.1 Product Perspective

Synsera is a web-based, AI-driven employee sentiment and burnout analysis system designed to help organizations monitor employee well-being and productivity. The system is largely ***self-contained*** *but can optionally interact with external organizational tools such as communication platforms or data storage systems.*

Unlike traditional employee feedback systems that rely on manual surveys and periodic reviews, Synsera provides ***continuous sentiment monitoring*** *through text-based employee check-ins. Similar products in the market mainly focus on performance metrics or survey-based analytics, whereas Synsera differentiates itself by emphasizing* ***emotional intelligence, NLP-based sentiment analysis, and early burnout detection****.*

Synsera is treated as a ***black-box system*** *from the user's perspective, where users interact through a web interface while internal AI processing remains hidden. The system may later be integrated with third-party platforms, but such integrations are not mandatory for its core functionality.*

2.1.1 System Interfaces

Synsera may interact with external systems such as:

- ***Organizational authentication systems*** *(for user login)*
- ***Communication platforms*** *(for future notification features)*
- ***External databases or analytics tools*** *(optional)*

Currently, the system primarily interfaces through web-based APIs to accept employee input and return sentiment insights. No mandatory dependency on existing enterprise systems is assumed at this stage.

2.1.2 User Interfaces

The system provides a ***Graphical User Interface (GUI)*** *accessible through a web browser. Employees submit daily or periodic check-ins through simple text forms, while managers and HR personnel access dashboards showing sentiment summaries and trends.*

The interface is designed to be:

- *Simple and intuitive*
- *Minimal in technical complexity*
- *Accessible to users with basic computer literacy*

Special care is taken to ensure readability, clarity, and ease of navigation, considering users from diverse educational and professional backgrounds.

2.1.3 Hardware Interfaces

The Synsera system does not directly interact with specialized hardware components. It runs on standard computing devices such as desktops, laptops, tablets, and smartphones through a web browser.

The system has no hardware interface requirements.

2.1.4 Software Interfaces

Table 2.1.4 System Interfaces

Name	Mnemonic	Version	Source
Web Browser	WB	Latest	User System
Database System	DBMS	MySQL 8.x	Open Source
NLP Libraries	NLP	Latest Stable	Open Source
Web Server	WS	Latest	Open Source

The system communicates with the database using standard SQL queries and interacts with NLP libraries through predefined APIs for text processing and sentiment classification.

2.1.5 Communications Interfaces

Synsera uses standard internet communication protocols:

- ***HTTP/HTTPS*** *for client-server communication*
- ***REST-based APIs*** *for data exchange between frontend and backend*

No custom communication protocol is required.

2.1.6 Memory Constraints

There are no strict memory constraints imposed by the customer. The system is expected to run on modern systems with standard memory availability. Memory usage will scale based on the number of users and stored sentiment records.

2.1.7 Operations

The system supports the following operational modes:

- ***Interactive operation*** *during employee check-in submission*
- ***Continuous background analysis*** *of sentiment data*
- ***Periodic generation*** *of summaries and insights*

Backup and recovery of data are handled through regular database backup procedures.

2.1.8 Site Adaptation Requirements

Before deployment, the following site-specific requirements must be met:

- *Database setup and initialization*
- *User roles (Employee, Manager, HR) must be configured*
- *Network connectivity must be available*

No physical modifications to the site are required.

2.2 Product Functions

The **Synsera** system provides a set of integrated functions to support continuous monitoring of employee emotional well-being and productivity. The system is designed to serve three primary user groups: **employees**, **managers**, and **HR personnel**. The following is a summary of the major functions performed by the software, presented in a manner understandable to non-technical users.

Major Functional Areas

1. User Account Management

- Allows employees, managers, and HR users to register and log into the system
- Maintains secure user sessions
- Supports role-based access control and secure logout

2. Employee Check-in Submission

- Enables employees to submit regular text-based check-ins
- Allows users to describe current tasks, challenges, and emotional state
- Stores submitted data securely for further analysis

3. Automated Sentiment Analysis

- Analyzes employee check-ins using AI-based sentiment classification
- Categorizes inputs as positive, negative, or neutral
- Assigns sentiment scores for further trend analysis

4. Sentiment Trend and Burnout Analysis

- Tracks sentiment patterns over time
- Identifies declining emotional trends and burnout risks
- Generates summarized insights for management review

5. Dashboard and Visualization

- Displays sentiment summaries and trends through dashboards
- Enables managers and HR personnel to view analytics at team or organization level
- Presents insights in a clear and easy-to-understand format

6. Security and Access Control

- Ensures confidentiality of employee data
- Restricts access to sensitive information based on user roles
- Protects system data through authentication and authorization mechanisms

Functional Relationship Overview

Employee → Account Management → Check-in Submission → Sentiment Analysis

Manager → Account Management → Dashboard Viewing → Trend Analysis

HR → Account Management → Organization-wide Insights → Burnout Monitoring

The above representation shows the **logical relationship between major system functions** and is **not intended to describe the internal design or implementation** of the system.

2.3 User Characteristics

The intended users of Synsera include employees, team leads, and HR personnel. Users are expected to have basic computer literacy and familiarity with web-based applications. No advanced technical expertise is required.

Managers and HR users may have moderate experience with dashboards and analytical tools, which influences the need for clear visualization and simple data interpretation.

User Categories

1. Employees

- **Educational Level:** Undergraduate or equivalent professional qualification
- **Technical Expertise:** Basic computer and internet usage
- **Experience:** Familiar with web-based applications such as internal portals and online forms

2. Team Leads / Managers

- **Educational Level:** Undergraduate or postgraduate
- **Technical Expertise:** Moderate experience with web-based dashboards
- **Experience:** Familiar with reviewing reports, summaries, and performance metrics

3. HR Personnel

- **Educational Level:** Undergraduate or postgraduate
- **Technical Expertise:** Moderate analytical and system usage skills
- **Experience:** Familiar with organizational data analysis and reporting tools

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Impact on System Design

- The user interface must be simple, intuitive, and easy to navigate
- Text input forms should be minimal and clearly guided
- Dashboards must present insights using clear visualizations and summaries
- Analytical data should be easy to interpret without technical knowledge
- System interactions should require minimal training

These characteristics justify the use of a **graphical web-based interface**, role-based access control, and simplified data visualization, which are further detailed in later sections of this document.

2.4 Constraints

The **Synsera** system is subject to several constraints that may limit design or operational choices. These constraints are described below in customer-understandable terms.

1. Data Privacy and Confidentiality

- Employee data must be handled securely and confidentially
- Personal and emotional information should not be exposed to unauthorized users

2. Secure Data Storage

- Employee check-ins and sentiment records must be stored securely
- Data access should be restricted based on user roles

3. Organizational IT Policies

- The system must comply with internal IT and data usage policies of the organization
- Access control and authentication rules must align with organizational standards

4. Parallel Operation

- Multiple users may access the system simultaneously
- The system must handle concurrent employee submissions without data loss

5. Reliability Requirements

- The system should remain available during normal working hours
- Downtime should be limited to maintenance or hosting issues

6. Ethical Use of AI Insights

- AI-generated sentiment insights must be used responsibly
- The system should not be used for punitive or discriminatory decision-making

7. Control Functions

- Access to dashboards and analytics must be restricted to authorized roles
- Employees should only view their own submissions

8. Interface to Other Applications

- Future integration with organizational tools may be required
- Such integrations should not compromise data security

9. Safety and Security Considerations

- Protection against unauthorized system access
- Secure handling of login credentials and session data

2.5 Assumptions and Dependencies

The following assumptions and dependencies affect the requirements stated in this Software Requirements Specification (SRS).

Assumptions

- Users have access to a stable internet connection
- Users use supported and modern web browsers
- Employees provide text-based check-ins honestly and regularly

Dependencies

- Availability of web hosting and server infrastructure
- Reliability of database services used for storing employee data
- Proper configuration of system environments and security settings

Any changes in these assumptions or dependencies may require modifications to the system requirements.

2.6 Apportioning of Requirements

Due to limitations in development time, available resources, and project scope, not all desired requirements can be implemented in the initial version of the **Synsera** system. Therefore, the system requirements are divided into phases to support incremental development and future enhancements. This apportioning helps prioritize essential functionality while allowing advanced features to be deferred to later versions.

Phase-wise Requirement Distribution

Phase 1 – Core System (Implemented)

- User registration and secure login
- Role-based access control (Employee, Manager, HR)
- Employee text-based check-in submission
- Automated sentiment classification
- Sentiment trend analysis
- Burnout risk identification
- Dashboard-based visualization for managers and HR
- Secure storage and access of employee data

Phase 2 – Enhancements (Future Scope)

- Integration with third-party collaboration tools (e.g., Slack, Teams)
- Improved sentiment visualization and reporting
- Scheduled sentiment summaries and alerts

Phase 3 – Extended Features (Future Work)

- Voice-based sentiment and emotional tone analysis
- Advanced predictive analytics for burnout and performance forecasting
- AI-driven personalized recommendations for employee well-being

This phased approach supports an **iterative development lifecycle** and ensures that critical system requirements are delivered first, while providing a clear roadmap for future expansion.

2.7. Use case

Use cases describe how different users of the **Synsera** system interact with the software to accomplish specific objectives. They represent the functional requirements of the system from the user’s point of view and help in understanding how the system behaves in various scenarios. Use cases focus on *what* the system does in response to user actions rather than *how* the system is internally implemented. This approach ensures clarity for stakeholders and provides a clear basis for further system analysis and design.

2.7.1. Use case Model

The use case model represents the functional behavior of the **Synsera** system from the user’s perspective. It identifies the different actors interacting with the system and the use cases that describe the services provided by the system. The following guidelines have been followed while preparing the use case model for Synsera to ensure clarity and consistency.

Guidelines for Use Case Modeling

- Primary actors are placed in the **top-left corner** of the use case diagram
- All actors are drawn **outside the system boundary**
- Actors are named using **singular, business-relevant nouns** such as *Employee*, *Manager*, and *HR*
- Each actor is associated with **one or more use cases** representing system functionality
- Actors represent **roles**, not individual job positions
- The stereotype **<<system>>** is used to indicate system actors, where applicable
- Actors do **not interact directly** with one another within the use case diagram
- An actor named **“Time”** is introduced to initiate scheduled or automated events
- Associations are represented using **lines connecting actors and use cases**, with optional open-headed arrowheads to indicate the direction of interaction
- Generalization relationships are depicted using **closed-headed arrows**, pointing toward the more general actor or use case

These guidelines ensure that the use case diagram clearly communicates system functionality while remaining independent of internal implementation details.

2.7.2 Use Case Diagram

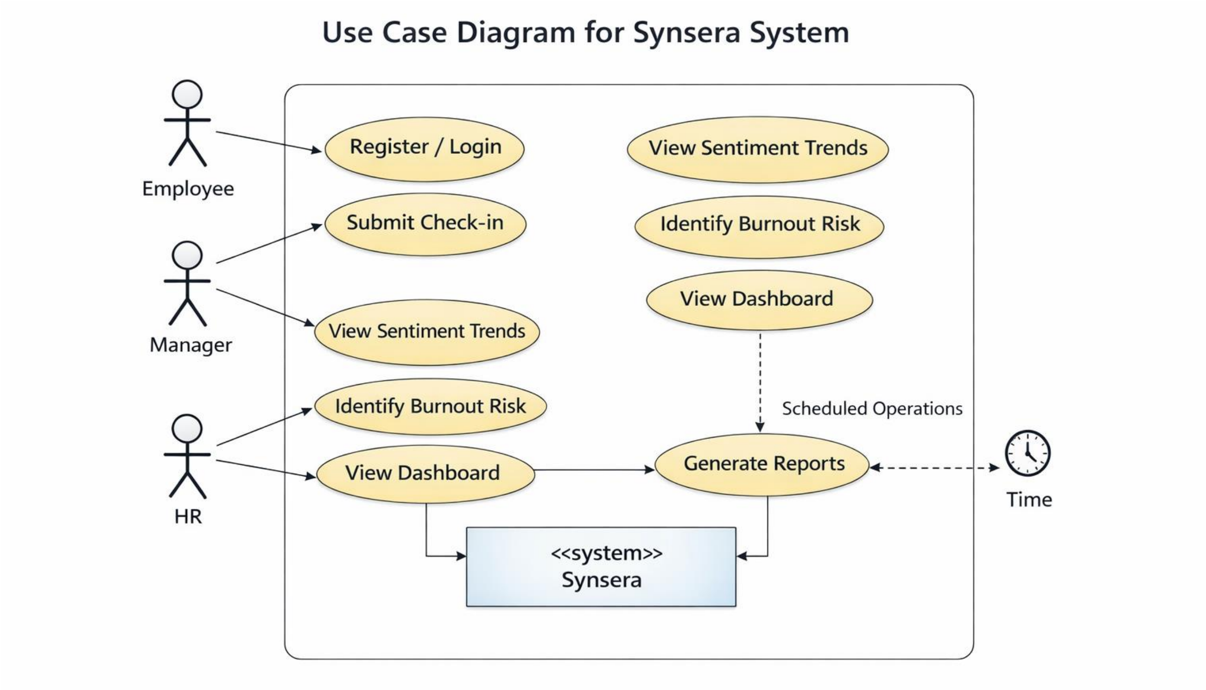

Figure 2.7.2. Use Case Diagram for Synsera

2.7.3 Use Case Scenario

Use Case 1: Employee Check-in Submission

Use Case Element	Description
Use Case Number	UC-01
Application	Synsera
Use Case Name	Submit Employee Check-in
Use Case Description	Allows an employee to submit a text-based work and mood update
Primary Actor	Employee
Precondition	Employee is registered and logged into the system
Trigger	Employee selects "Submit Check-in" option

Basic Flow

1. Employee logs into the system
2. Employee opens the check-in form
3. Employee enters text describing tasks and feelings
4. Employee submits the check-in
5. System stores the data and analyzes sentiment

Alternate Flows

1. If the network connection fails, submission is delayed
2. If input is empty, system prompts the user to enter valid text

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Use Case 2: View Sentiment Dashboard

Use Case Element	Description
Use Case Number	UC-02
Application	Synsera
Use Case Name	View Sentiment Dashboard
Use Case Description	Allows managers and HR to view sentiment trends and summaries
Primary Actor	Manager / HR
Precondition	User is authenticated with appropriate role
Trigger	User selects “View Dashboard” option

Basic Flow

1. User logs into the system
2. User navigates to the dashboard page
3. System retrieves sentiment data
4. Dashboard displays trends and summaries

Alternate Flows

1. If data is unavailable, system displays an informational message

2. If access is unauthorized, system denies access

Use Case 3: Generate Sentiment Report

Use Case Element	Description
Use Case Number	UC-03
Application	Synsera
Use Case Name	Generate Sentiment Report
Use Case Description	Automatically generates sentiment reports at scheduled intervals
Primary Actor	Time
Precondition	Sentiment data exists in the system
Trigger	Scheduled system time event

Basic Flow

1. Time triggers scheduled operation

2. System collects sentiment data

3. System generates summary report

4. Report is stored and made available to HR

Alternate Flows

1. If data is insufficient, system generates a partial report

2. If system is unavailable, report generation is postponed

2.8 Sequence diagrams

2.8.1 User Registration Process

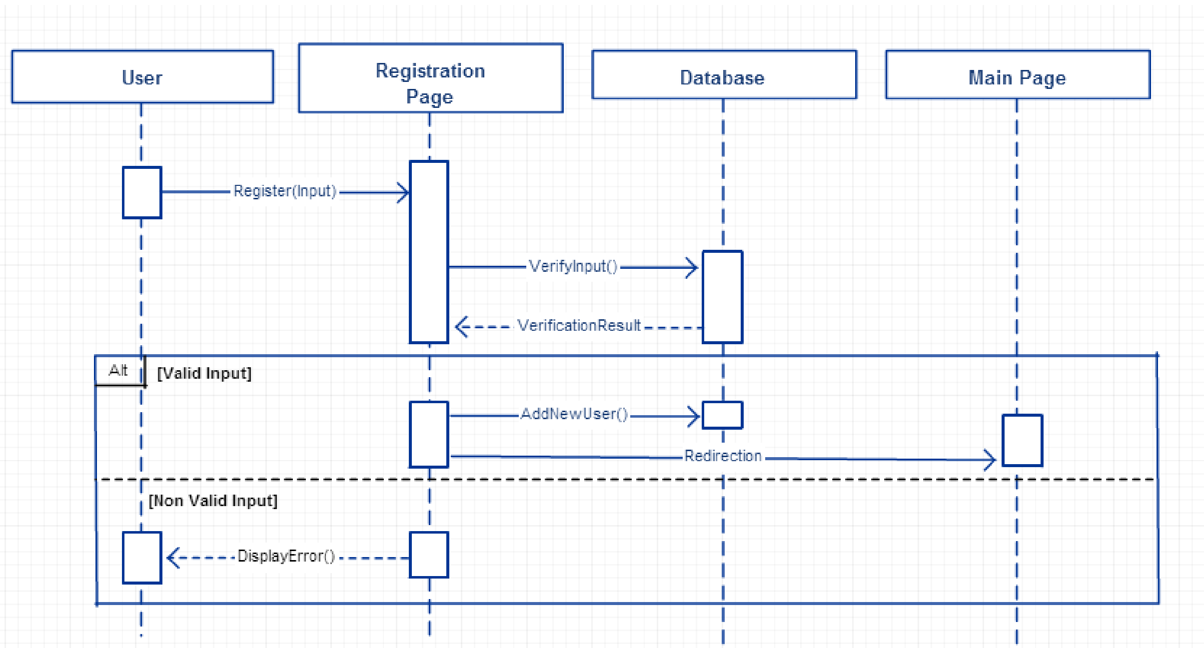

Figure 2.8.1 Sequence diagram for user registration process

CHAPTER 3
SYSTEM DESIGN

3.1.Architecture diagrams

The **Synsera** system follows a **three-tier architecture**, which is a widely adopted architecture for modern web-based and AI-driven applications. This architectural approach separates the system into well-defined layers, thereby improving **scalability, maintainability, flexibility, and security**. Each layer has a specific responsibility and interacts with adjacent layers through well-defined interfaces.

The three-tier architecture ensures that changes in one layer (such as the user interface or data storage) do not directly affect other layers, making the system easier to extend and maintain.

Architecture Layers

1. Presentation Layer

- Provides a web-based interface for employees, managers, and HR users.
- Handles user interaction, check-in submission, and dashboard visualization.

2. Application Layer

- Contains the core business logic of the system.
- Processes sentiment analysis, authentication, and burnout detection.
- Acts as a bridge between the presentation and data layers.

3. Data Layer

- Responsible for storing and retrieving persistent data
- Maintains user profiles, employee check-ins, sentiment scores, and historical trends
- Ensures secure and reliable data access for analytics and reporting

External services such as **NLP libraries**, **machine learning models**, and optional **notification services** interact with the application layer to provide intelligent sentiment analysis and reporting functionality.

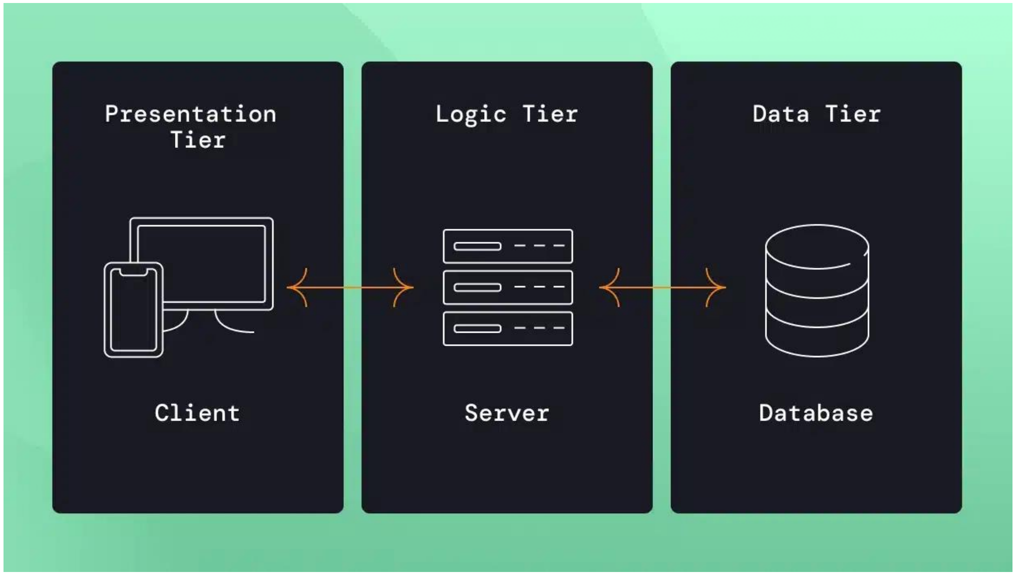

Figure 3.1 Three-Tier Architecture Diagram of Synsera

3.2.Class diagrams

The class diagram represents the **static structure of the Synsera system**. It illustrates the main system classes, their attributes, and the relationships between them. This diagram helps in understanding how different entities involved in employee sentiment analysis are logically organized and how they interact with each other.

Major Classes Identified

- **User**
    - userId, name, email, password, role
- **EmployeeCheckIn**
    - checkInId, message, timestamp, sentimentScore
- **SentimentAnalysis**
    - analysisId, sentimentType, confidenceScore
- **Dashboard**
    - dashboardId, summaryData, trendData

- **Report**
    - reportId, generatedDate, reportContent

Relationships

- One **User** can submit multiple **EmployeeCheckIn** entries
- Each **EmployeeCheckIn** is associated with one **SentimentAnalysis** result
- **Dashboard** aggregates sentiment data from multiple users
- **Report** is generated based on dashboard analytics and sentiment trends

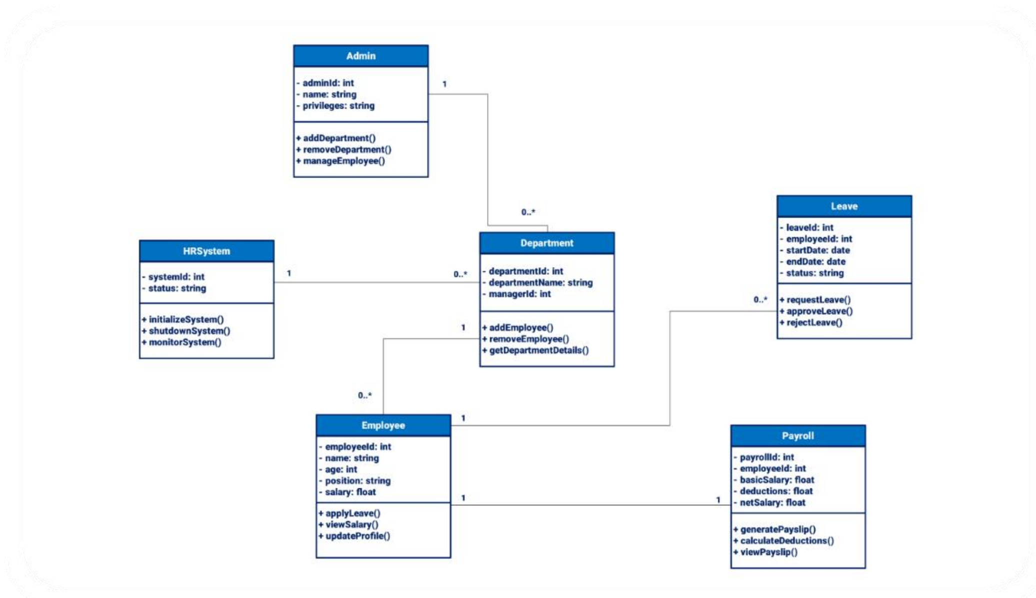

Figure 3.2: Class Diagram of Synsera

3.3. Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how data moves through the **Synsera system**. It identifies the main processes, data stores, and external entities involved in system operation. The DFD helps in understanding how employee inputs are processed and transformed into meaningful sentiment insights without focusing on implementation details.

Level 0 DFD (Context Diagram)

At Level 0, the entire Synsera system is represented as a single process interacting with external entities.

- Employees provide inputs such as login credentials and text-based check-ins
- Managers and HR users request dashboards and reports
- The system processes requests and performs sentiment analysis
- Data is stored in and retrieved from the central database

Level 1 DFD

At Level 1, the Synsera system is decomposed into major internal processes. Each process interacts with the database and external entities as required.

- **User Authentication** – validates user credentials and manages access
- **Employee Check-in Processing** – accepts and stores employee text inputs
- **Sentiment Analysis** – analyzes check-ins and generates sentiment scores
- **Dashboard & Reporting** – presents sentiment trends and burnout indicators to managers and HR

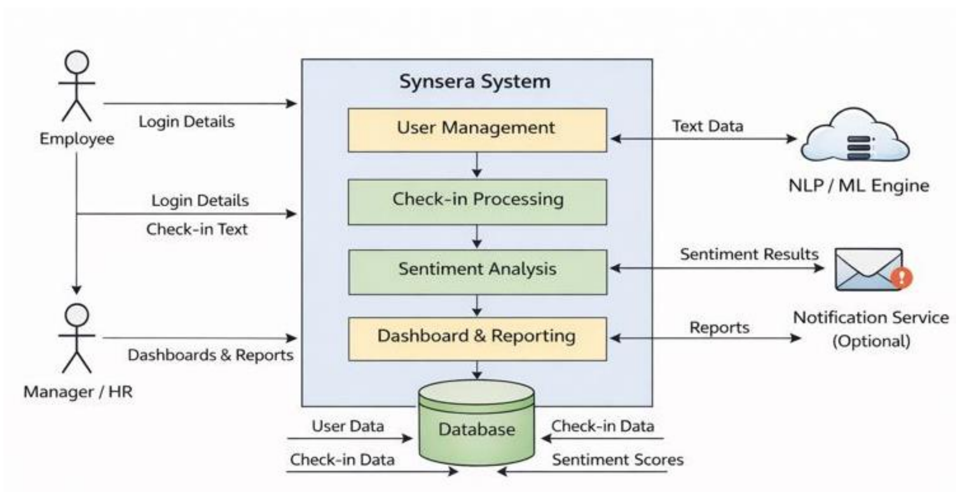

Figure 3.3: Data Flow Diagram of Synsera

3.4. Activity Diagram

The activity diagram represents the **workflow of activities** performed during **user registration and login** in the Synsera system. It illustrates the sequence of actions, decision points, and system responses, helping to understand how users interact with the system at a process level.

Registration Activity Flow

- User enters registration details
- System validates the provided information
- A new user account is created
- Confirmation message is displayed to the user

Login Activity Flow

- User enters login credentials
- System verifies the credentials
- User is authenticated successfully
- Access is granted to the user dashboard

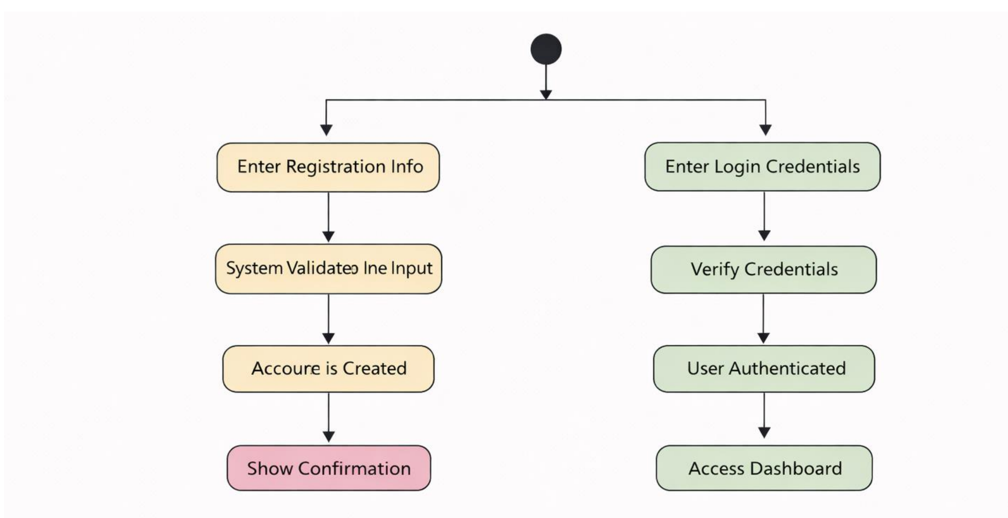

Figure 3.4: Activity Diagram for User Registration and Login in Synsera

3.5.ER Diagrams

The Entity Relationship (ER) diagram illustrates the **logical structure of the Synsera database**. It shows the major entities, their attributes, and the relationships between them. This diagram helps in understanding how data related to employee sentiment analysis is organized and stored within the system.

Entities Identified

- User
- EmployeeCheckIn
- SentimentAnalysis
- Report
- Dashboard

Relationshis

- A **User** submits multiple **EmployeeCheckIn** records
- Each **EmployeeCheckIn** is associated with one **SentimentAnalysis** result
- **Dashboard** aggregates sentiment data from multiple users

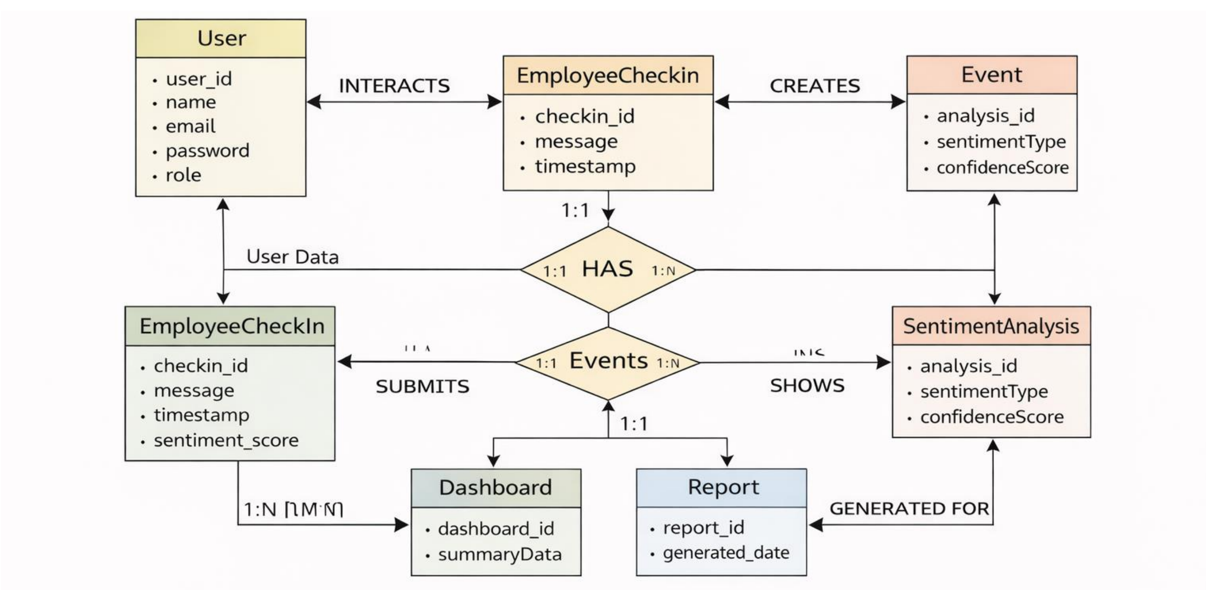

Figure 3.5: Entity Relationship Diagram of Synsera

3.6.Database schema diagrams

The database schema diagram represents the structure of the database tables and their fields used in the **Synsera system**. It defines how data is stored and organized to support sentiment analysis and reporting.

Schema Overview

- **Users Table:** Stores user credentials and role information
- **EmployeeCheckIn Table:** Stores employee check-in messages and timestamps
- **SentimentAnalysis Table:** Stores sentiment analysis results
- **Report Table:** Stores generated sentiment reports

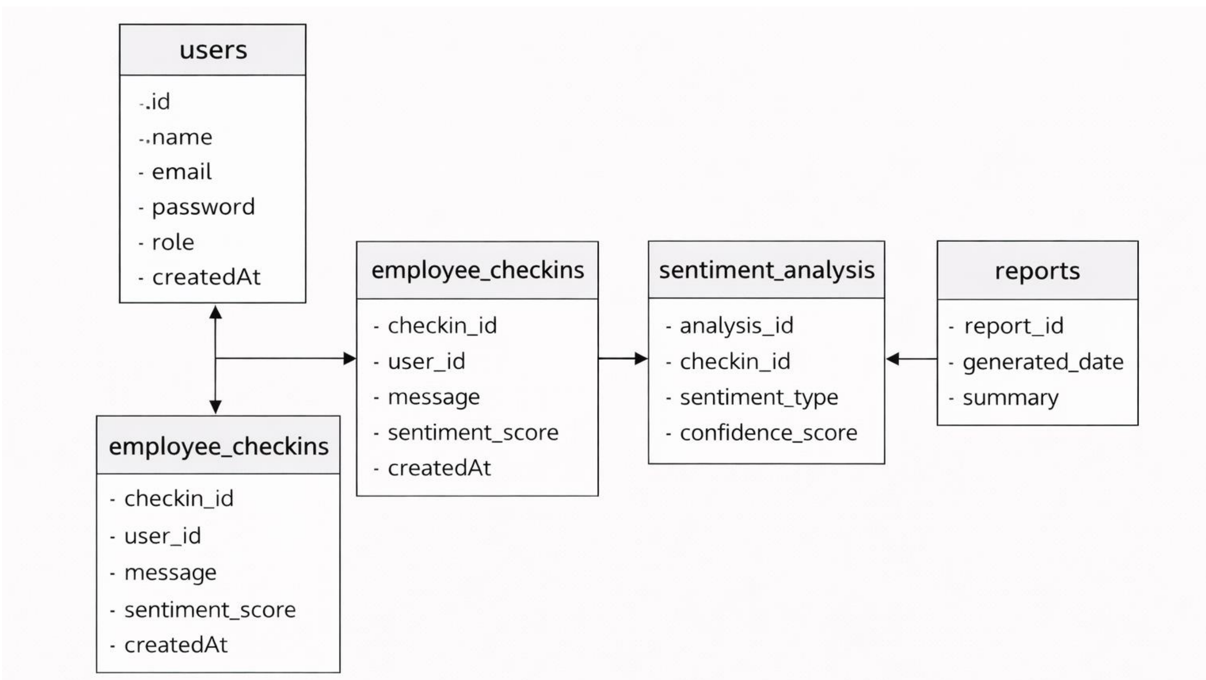

Figure 3.6: Database Schema Diagram of Synsera

CHAPTER 4

IMPLEMENTATION AND RESULTS

This chapter describes the implementation details of the **Synsera** web-based application and the results obtained after testing the system. It outlines the software and hardware requirements considered during development, the assumptions and constraints involved, the implementation approach adopted, and the test cases executed to validate system functionality. The chapter concludes by presenting the final outcomes achieved through the successful implementation of the employee sentiment and burnout analysis system.

4.1. Software and Hardware Requirements

This section describes the software and hardware resources required for the successful implementation and execution of the Synsera system.

Table 4.1 Software Requirements

Software Component	Description
Operating System	Windows / Linux / macOS
Frontend	React.js with Vite
Backend	Node.js with Express.js
Database	MongoDB Atlas (Cloud)
Styling	Tailwind CSS
API Communication	Axios
Authentication	JWT with HTTP-only Cookies
Image Storage	Cloudinary
Email Service	SMTP (Brevo)
Deployment	Vercel (Frontend), Render (Backend)
Browser	Chrome, Firefox, Edge

Table 4.2 Hardware Requirements

Hardware Component	Minimum Requirement
Processor	Dual Core Processor
RAM	4 GB
Storage	10 GB Free Disk Space
Internet	Stable Internet Connection

4.2. Assumptions and dependencies

The implementation of the **Synsera** system is based on the following assumptions and dependencies:

Assumptions

- Users have access to a modern web browser
- Stable internet connectivity is available during system usage
- Users possess basic knowledge of web-based applications
- Cloud services required by the system remain operational

Dependencies

- Availability of **MongoDB Atlas** for data storage
- Availability of **NLP and machine learning libraries** for sentiment analysis
- Availability of **SMTP service** for email notifications (if enabled)
- Hosting platforms (**Vercel** for frontend and **Render** for backend) functioning correctly

Any changes in these assumptions or dependencies may impact the functionality and performance of the system.

4.3. Constraints

During the development and deployment of the **Synsera** system, the following constraints were identified:

- Use of free-tier cloud hosting may result in cold-start delays
- Limited computational resources available on free hosting plans
- Real-time sentiment processing and notifications are not implemented
- Dependency on third-party services for database hosting, NLP libraries, and email delivery

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4.4. Implementation Details

The **Synsera** system was implemented using a **modular and layered approach**, with a clear separation between frontend and backend responsibilities. The frontend is responsible for user interaction, data input, and visualization of sentiment dashboards, while the backend manages business logic, sentiment analysis processing, authentication, and database operations.

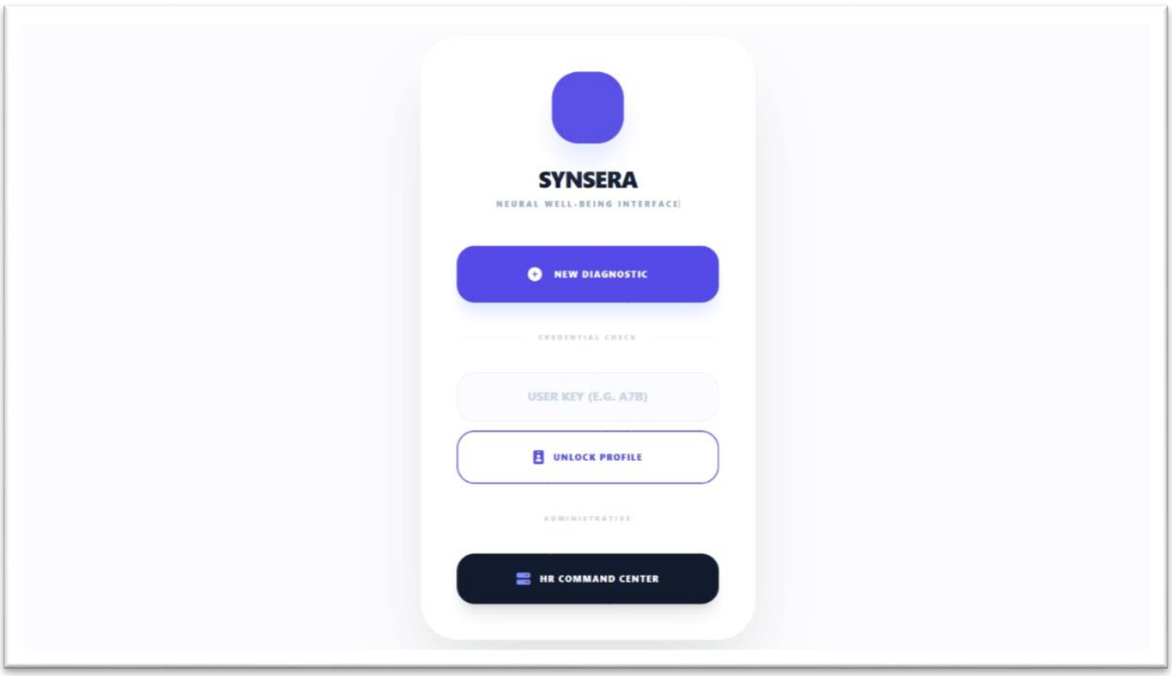
The system supports three primary user roles:

- **Employee:** Can submit text-based check-ins and view personal sentiment history
- **Manager:** Can view team-level sentiment trends and burnout indicators
- **HR:** Can access organization-wide analytics and reports

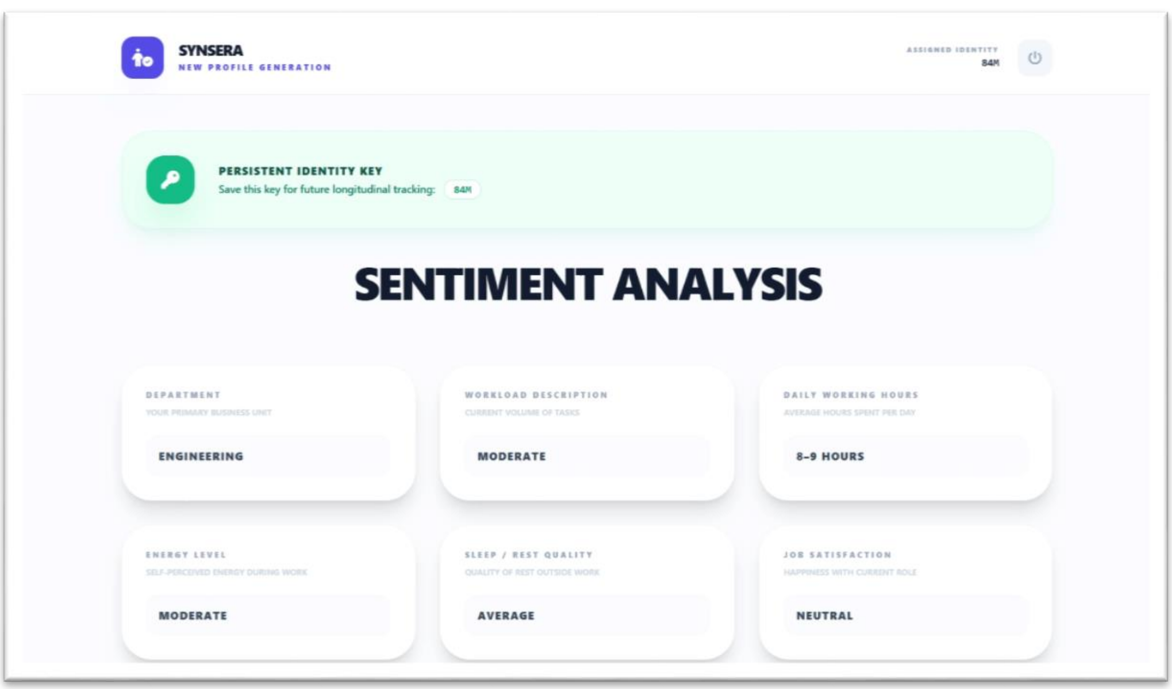
Security mechanisms such as **JWT-based authentication**, **HTTP-only cookies**, **role-based access control**, and **request validation** were implemented to ensure data privacy and protect sensitive employee information.

4.4.1 Snapshots Of Interfaces

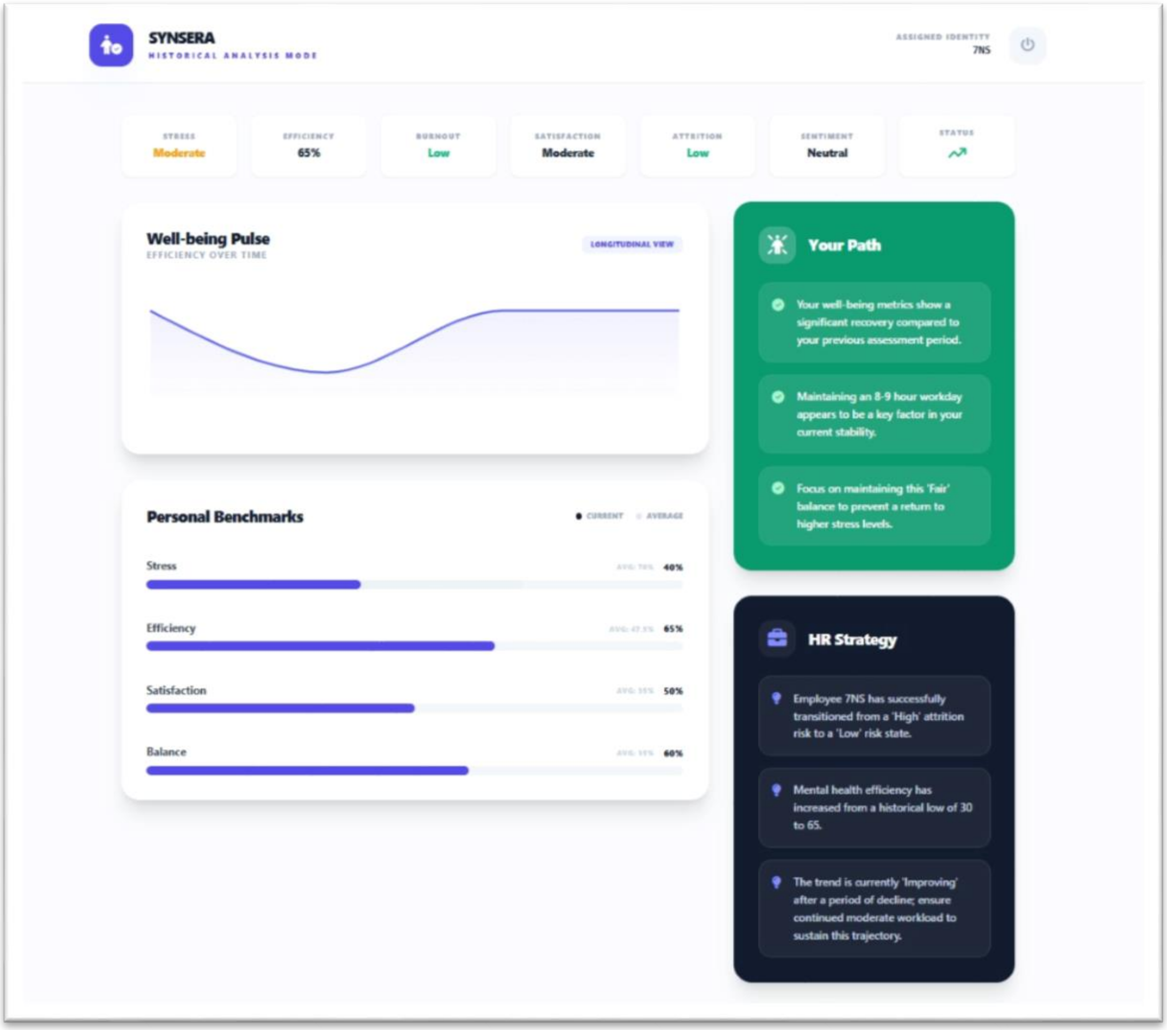
Screenshot 1: New Profile Generation / User Identity Creation



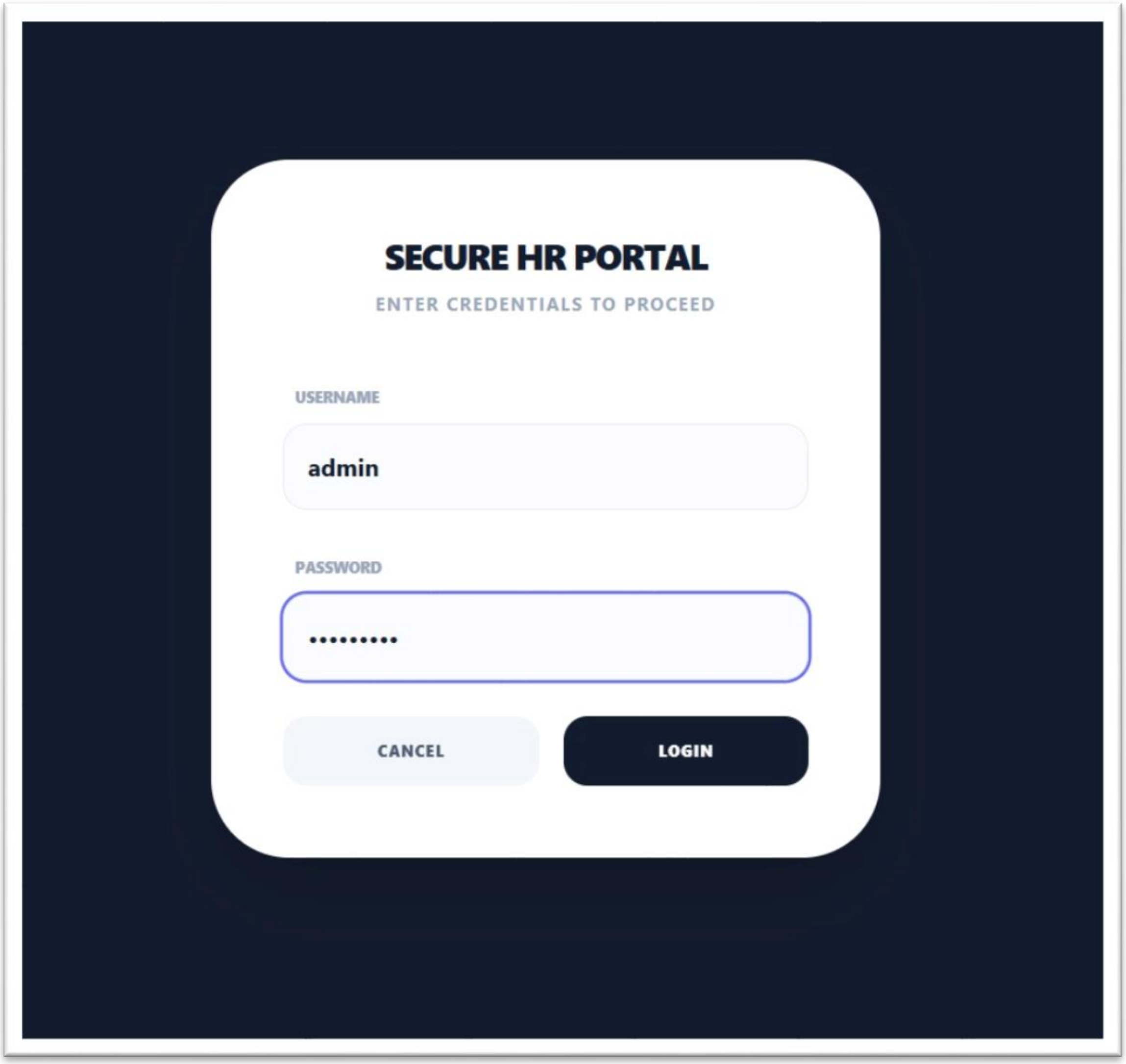
Screenshot 2: Sentiment Input & Self-Assessment Screen



Screenshot 3: Individual Sentiment Analysis & Well-Being Dashboard



Screenshot 4: Secure HR Login



Screenshot 5: HR Admin Login & Organizational Analytics Dashboard

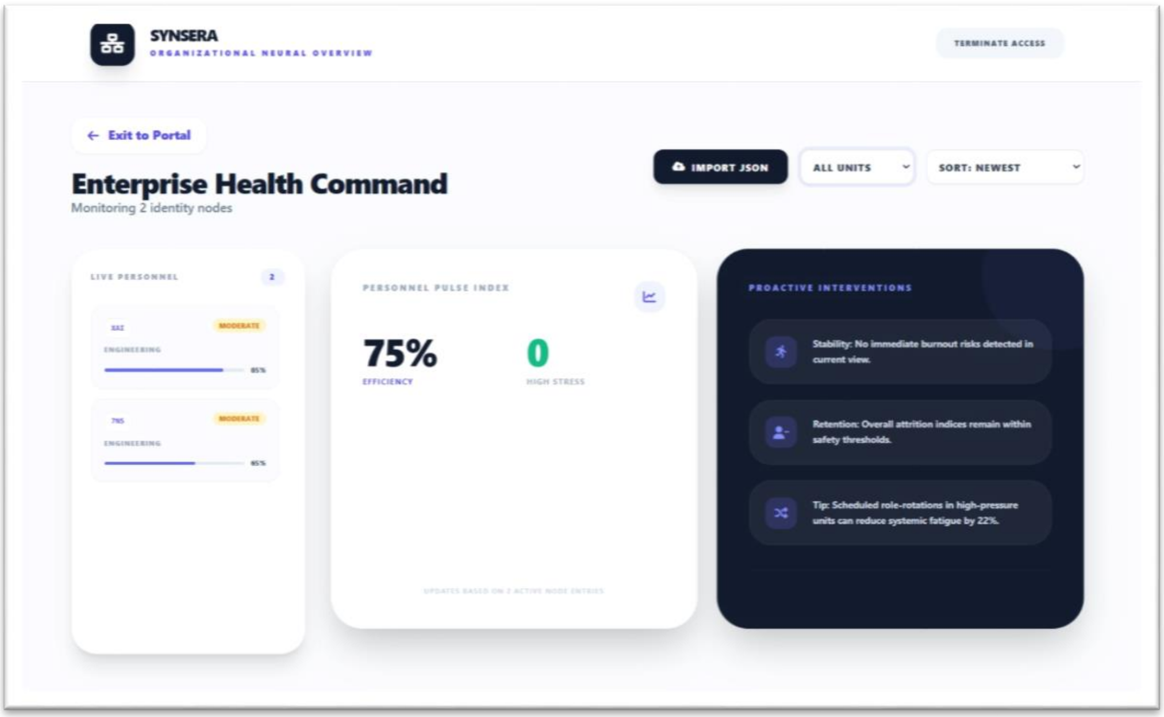


Figure 4.1 – Figure 4.5: User Interface Screenshots

4.4.2 Test Cases

Table 4.4.2 Test Cases

Test Case ID	Description	Input	Expected Output	Result
TC-01	User Profile Creation	Valid user details	Profile created successfully	Pass
TC-02	User Login	Valid credentials	Login successful	Pass
TC-03	Invalid Login	Incorrect user key / credentials	Error message displayed	Pass
TC-04	Submit Check-in	Valid text input	Check-in recorded	Pass
TC-05	Sentiment Analysis	Submitted check-in text	Sentiment classified correctly	Pass

TC-06	View Personal Dashboard	Authenticated user	Sentiment dashboard displayed	Pass
TC-07	HR Dashboard Access	HR credentials	Organizational dashboard loaded	Pass
TC-08	Unauthorized Access	Direct URL access without login	Access denied	Pass

4.4.3 Results

The Synsera system was successfully implemented and tested. All core functionalities operated as expected under normal usage conditions.

Observed Results:

- Secure user authentication and role-based access control
- Successful submission and processing of employee check-ins
- Accurate sentiment classification and trend analysis
- Proper storage and retrieval of user and sentiment data
- Effective visualization of individual and organizational dashboards

The system achieved its intended objectives and demonstrated the feasibility of an AI-assisted, cloud-deployed employee sentiment and well-being monitoring platform. No direct comparison with commercial systems was performed; however, the project satisfies the functional requirements defined during the planning and design phases.

CHAPTER 5

CONCLUSION

This chapter concludes the **Synsera** project by evaluating the system's performance, comparing it with existing state-of-the-art approaches, and discussing potential future directions. The objective of this chapter is to reflect on the outcomes of the project, highlight its practical implications in organizational well-being monitoring, and identify opportunities for further enhancement. The discussion is presented in a clear and non-technical manner to ensure accessibility to a broad audience.

5.1 Performance Evaluation

The performance of the **Synsera** system was evaluated based on functionality, reliability, usability, and responsiveness. During testing, the system successfully handled core operations such as user authentication, employee check-in submission, sentiment analysis processing, and dashboard visualization without functional errors.

Since the application is deployed on cloud infrastructure, system performance is influenced by network connectivity and hosting service availability. Under normal operating conditions, the system responded efficiently to user requests. Minor delays were observed during initial access due to free-tier hosting limitations; however, these delays did not significantly affect system correctness or overall user experience.

Overall, Synsera meets its intended performance expectations within the defined scope of an academic project and demonstrates stable and reliable operation.

5.2 Comparison with Existing State-of-the-Art Technologies

Existing employee monitoring and sentiment analysis solutions range from basic survey-based tools to advanced commercial platforms offering real-time analytics and AI-driven insights. Compared to traditional manual surveys and periodic feedback mechanisms, **Synsera** provides a more automated, continuous, and data-driven approach to understanding employee sentiment.

While commercial systems may include advanced features such as real-time alerts, large-scale predictive analytics, and enterprise integrations, Synsera focuses on essential sentiment analysis functionality combined with privacy-aware design and modern web technologies. The project demonstrates that effective sentiment monitoring systems can be developed using contemporary NLP techniques and cloud services, even within academic and resource-constrained environments.

Thus, Synsera aligns well with current technological standards while maintaining simplicity, accessibility, and ethical considerations.

5.3 Future Directions

The Synsera project suggests several interesting avenues for future enhancement and research. Future work could include the integration of voice-based sentiment analysis, real-time sentiment alerts, and more advanced predictive burnout models using deep learning techniques. Integration with third-party workplace collaboration tools and mobile application support could further improve system usability and adoption.

From a practical perspective, Synsera has significant implications for organizations seeking non-intrusive methods to monitor employee well-being and productivity trends. The system can be adapted for use in corporate environments, educational institutions, and remote work settings to support data-informed HR decision-making.

The work carried out in this project provides a strong foundation for future development and highlights the importance of AI-assisted, secure, and well-structured systems in addressing real-world organizational challenges.

Appendix

The appendix includes supplementary information that supports the **Synsera** project but is not essential to the main flow of the report. This section is included to comply with academic documentation standards.

No additional material such as extended datasets, lengthy tables, or complete source code listings was required to be included as an appendix for this project. All important diagrams, test cases, system outputs, and results have been presented within the main chapters of the report.

The complete source code of the Synsera system is maintained separately in the project repository and follows proper modular structure, indentation, and documentation practices.

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